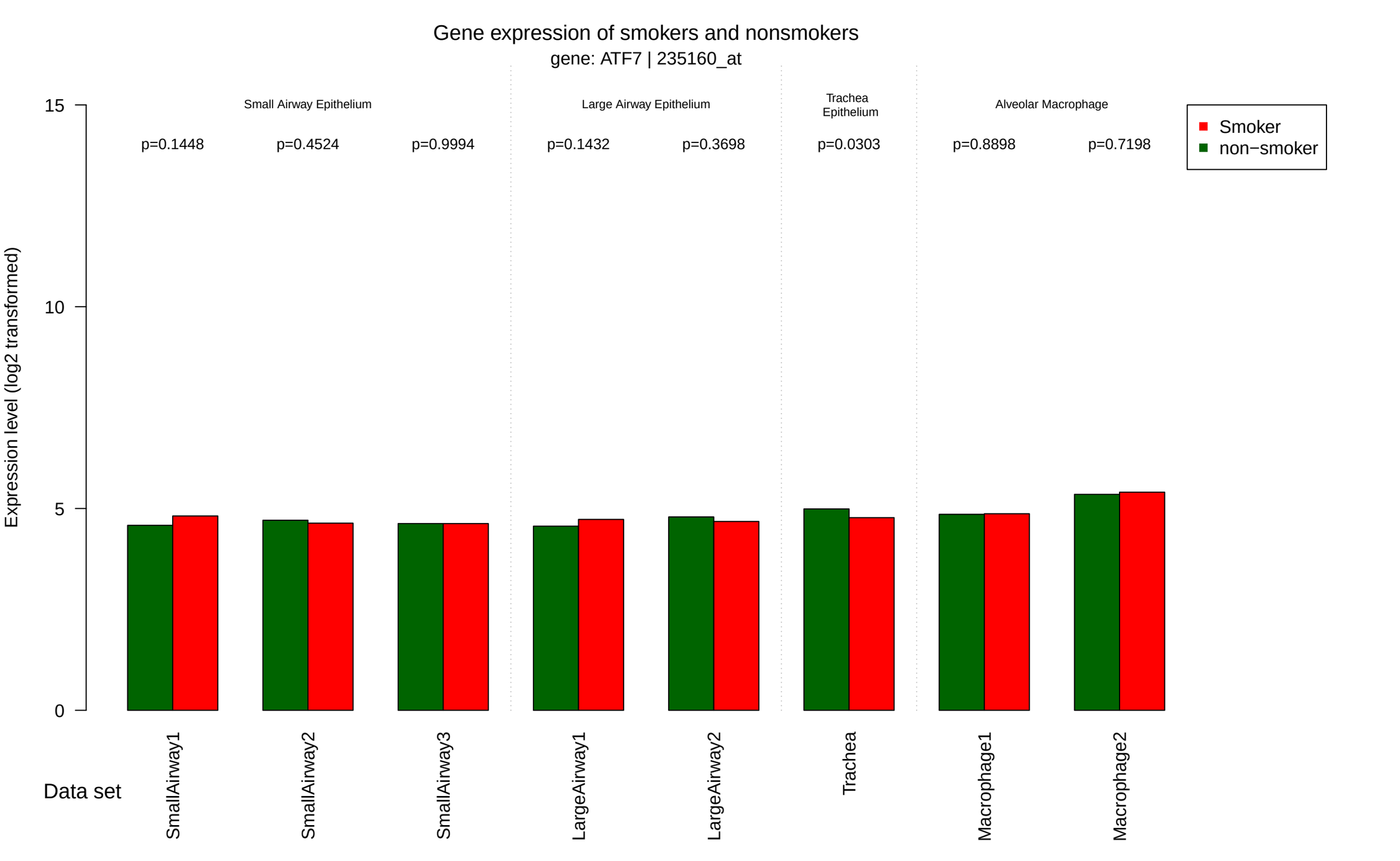
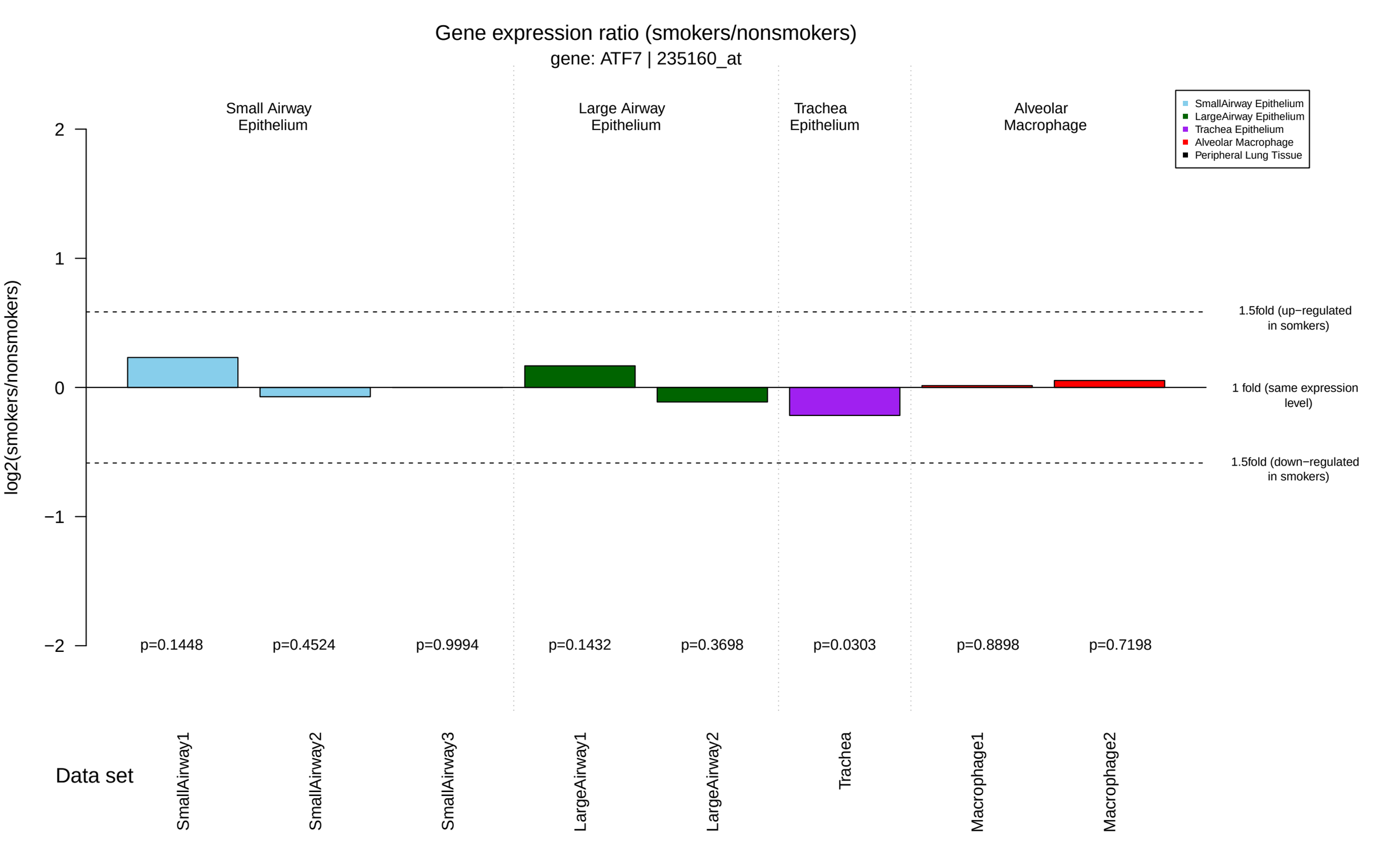
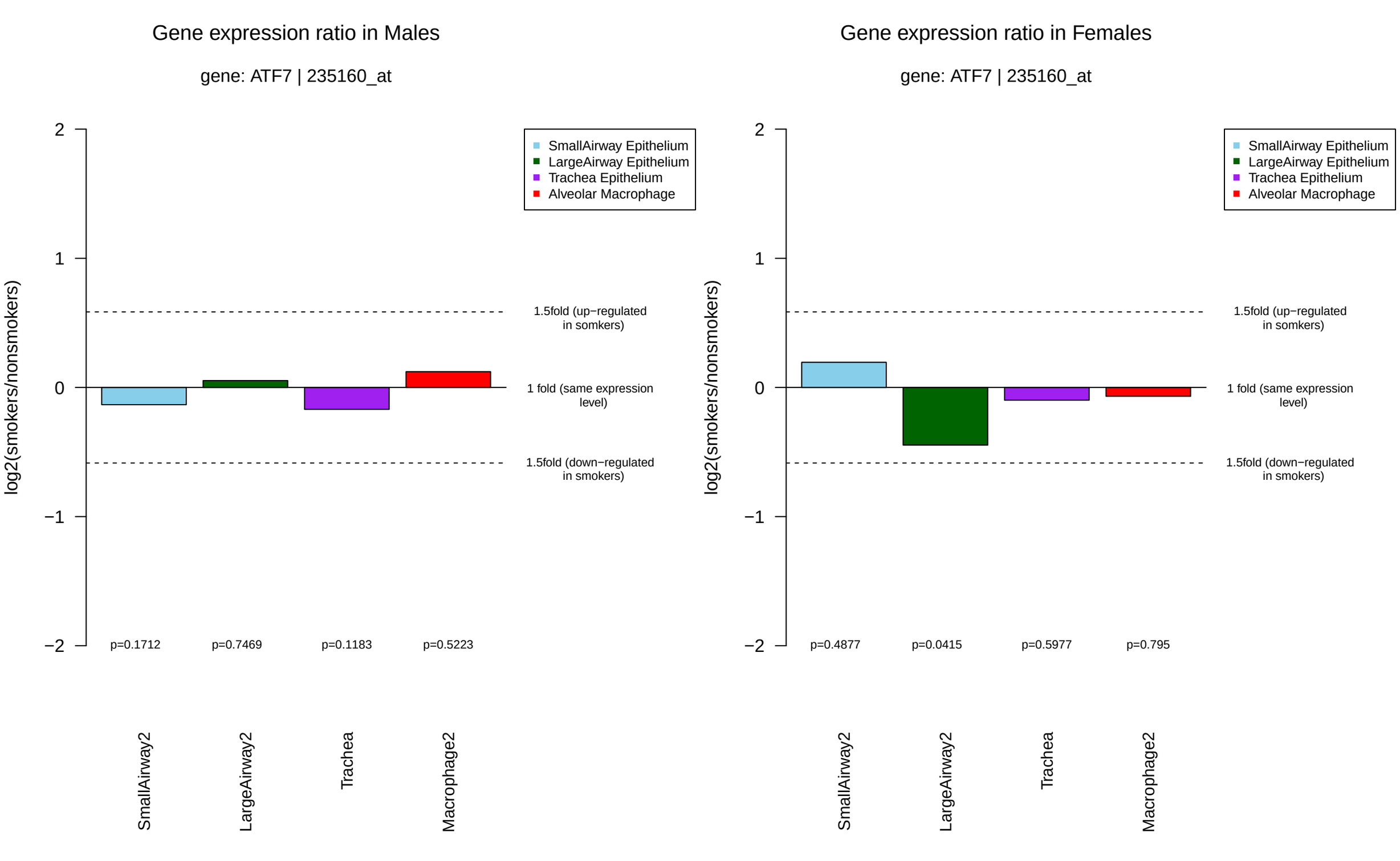
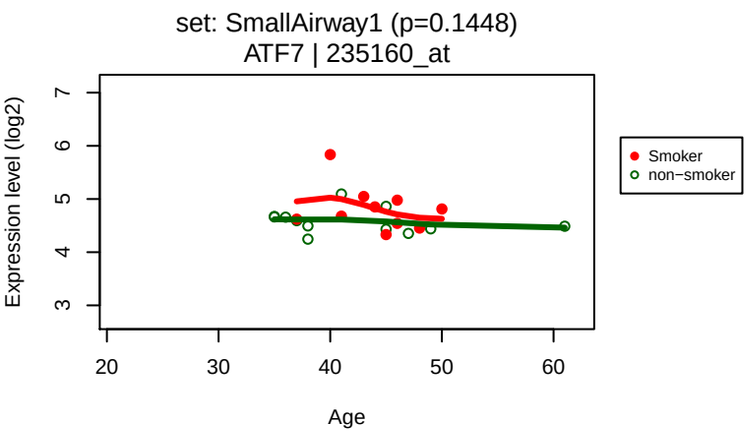
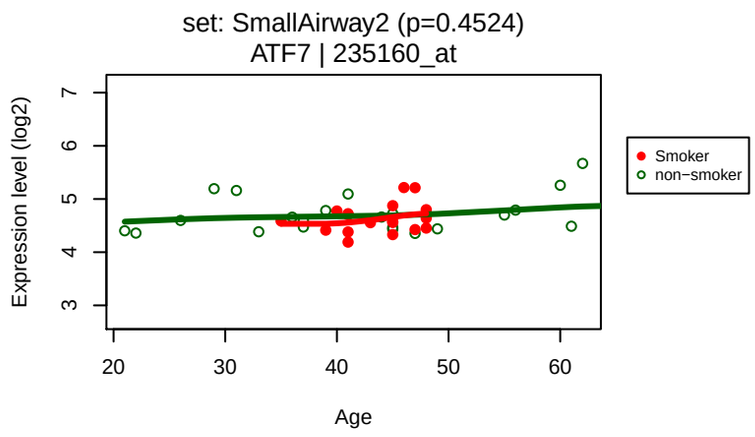
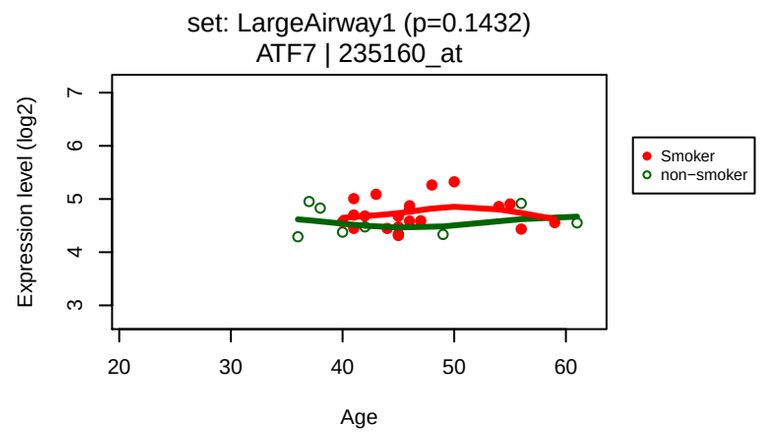

Gene expression of smokers and nonsmokers
gene: ATF7 | 235160_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1448
p=0.4524
p=0.9994
p=0.1432
p=0.3698
p=0.0303
p=0.8898
p=0.7198
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

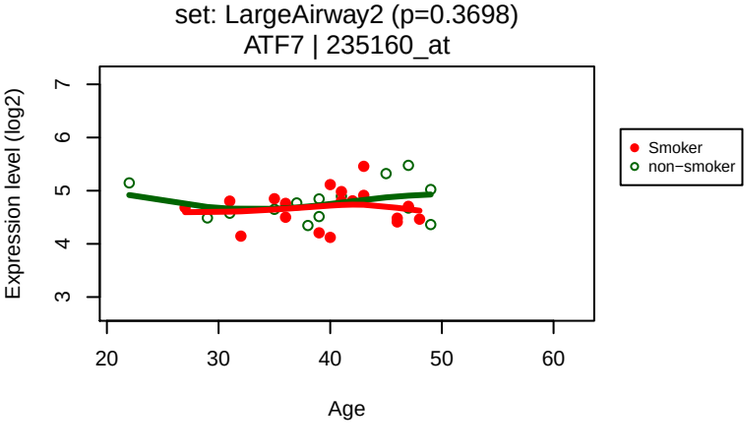

Gene expression ratio (smokers/nonsmokers)
gene: ATF7 | 235160_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1448
p=0.4524
p=0.9994
p=0.1432
p=0.3698
p=0.0303
p=0.8898
p=0.7198
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

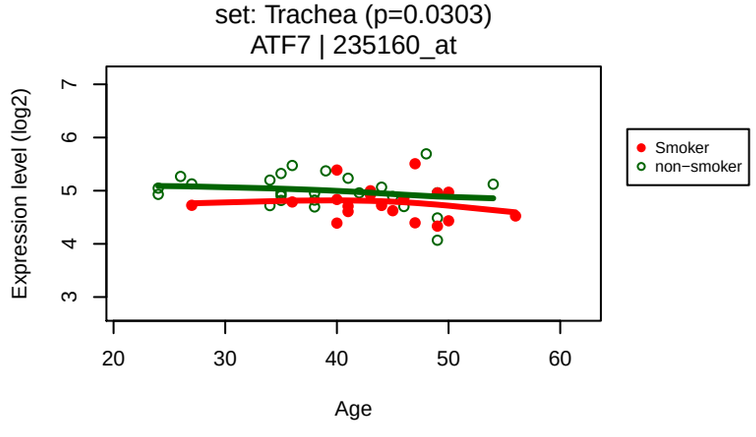

Gene expression ratio in Males
gene: ATF7 | 235160_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1712
p=0.7469
p=0.1183
p=0.5223
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF7 | 235160_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4877
p=0.0415
p=0.5977
p=0.795
SmallAirway2
LargeAirway2
Trachea
Macrophage2

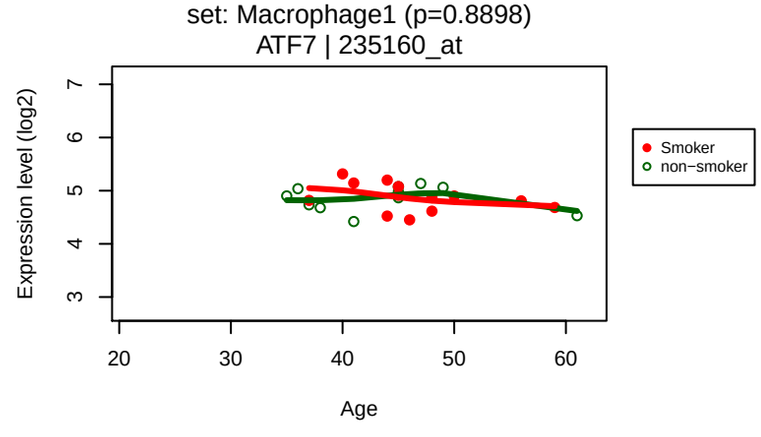

set: SmallAirway1 (p=0.1448)
ATF7 | 235160_at
Expression level (log2)
Age
Smoker
non−smoker

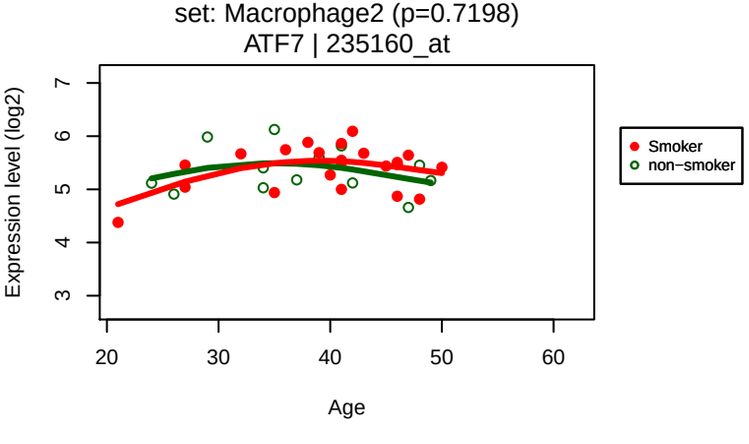

set: SmallAirway2 (p=0.4524)
ATF7 | 235160_at
Expression level (log2)
Age
Smoker
non−smoker

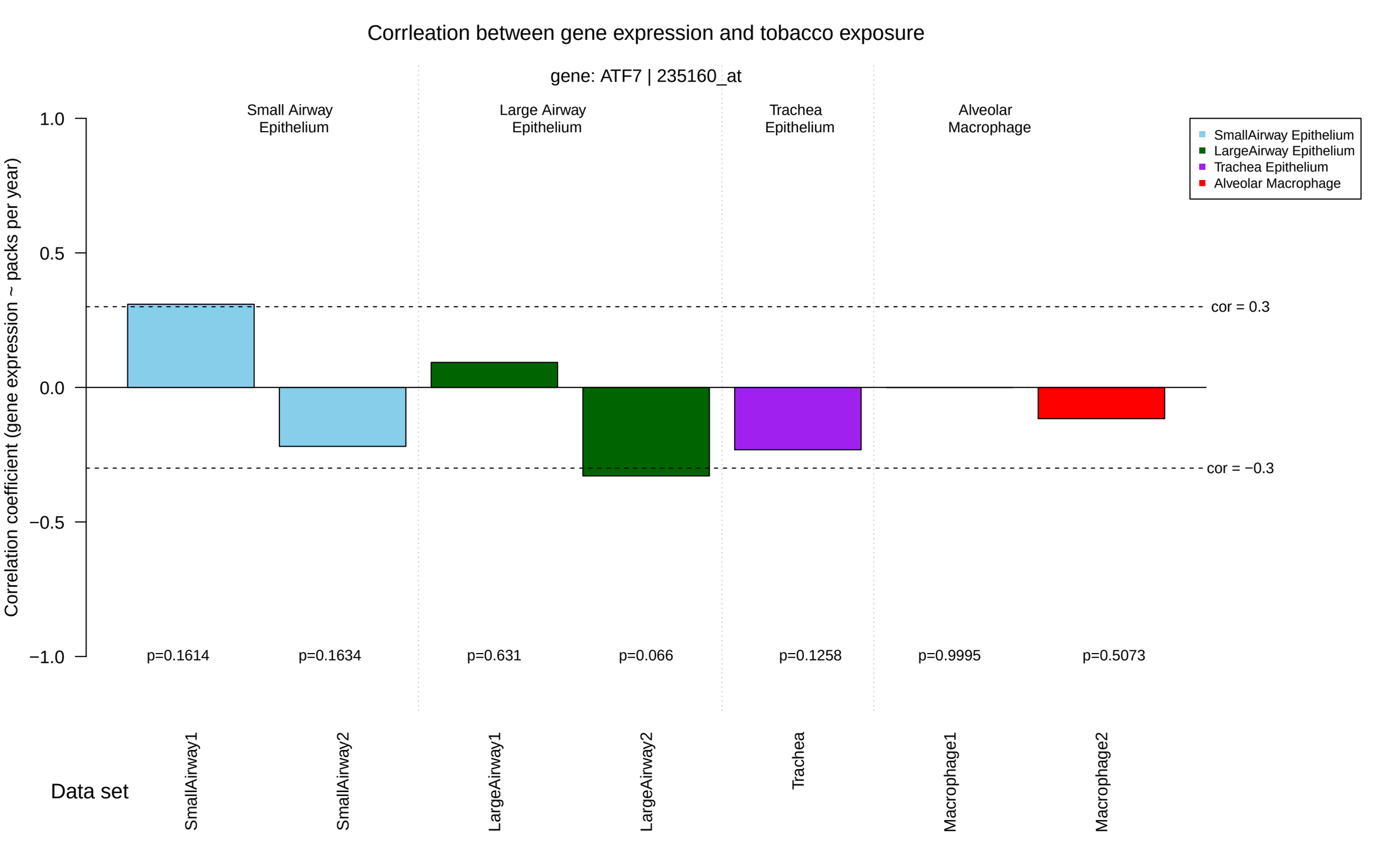

set: LargeAirway1 (p=0.1432)
ATF7 | 235160_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3698)
ATF7 | 235160_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0303)
ATF7 | 235160_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8898)
ATF7 | 235160_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7198)
ATF7 | 235160_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF7 | 235160_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1614
p=0.1634
p=0.631
p=0.066
p=0.1258
p=0.9995
p=0.5073
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2